CIP B106 Mobile and IoT Forensics

Android 10 (Pixel 3) Evidence Case Study

Digital Forensic Examination Report

Course code	CIP-B106
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Evidence	Pixel 3 Android 10 training logical copy (25 extracted SQLite databases)

1.0 Executive summary

This report examines 25 SQLite databases supplied from the Pixel 3 Android 10 training logical copy (case CIPB106-Android10-C11_26_DFIT_17305). Only databases were supplied; build.prop, packages.xml, usagestats, Wi-Fi and user XML files were not, so no build, package-usage or Wi-Fi findings are made. The databases show a Google account plus twelve messaging and social accounts registered between 29 January and 14 February 2020, two Google Fi SIM profiles, and 94 Play Store app records. Device-local time was UTC-05:00. A 24-hour window (8 February 19:30 to 9 February 19:30 UTC) is reconstructed with 18 events from eight databases and five cross-source relationships (two fully corroborated), including a WhatsApp location share and a geotagged photograph about 9.5 m apart.

2.0 Evidence and method

2.1 Evidence register

Item	Description	Size / SHA-256
db.zip	Archive supplied by the examiner containing 25 database files in folders Task A to Task D	543,451 bytes 0ce7de66942ce4f4001 2b0efd6a2948fb9139c9c2 87e2659dcc0b43cf15a25f1
Task A (3 files)	accounts_ce.db, accounts_de.db, telephony.db	Per-file hashes: Appendix F
Task B (4 files)	googlesettings.db, gservices.db, localappstate.db, locksettings.db	Appendix F
Task C (5 files)	calendar.db, calllog.db, contacts2.db, mmssms.db, profile.db	Appendix F
Task D (13 files)	annotated_call_log.db, bugle_db, dialer.db, gmm_myplaces.db, gmm_storage.db, gmm_sync.db, gphotos-1.db, kikCoreDatabase.db, local_trash.db, msgstore.db, textnow_data.db, voicemail.db, wa.db	Appendix F

Integrity check. The brief gives an expected SHA-256 (ca6918ef...8aa00a) for the complete course package. The archive examined here is a 25-file extract, not that package, so the expected value cannot be compared and integrity of the complete image is not verified. As a baseline, SHA-256 values were recorded for the archive and for every extracted file at the time of extraction (Appendix F). All databases were opened read-only with SQLite's immutable flag, so no journal or WAL file could be created or merged. Note that the file in the archive is named accounts_ce.db whereas Appendix A of the brief names account_ce.db; the supplied name is used throughout.

2.1 Acquisition scope

The course image is a logical copy created after TWRP root access; it is not a bit-for-bit image. Data that was not copied (deleted records in unallocated space, files outside the copied set) cannot be assessed. The TWRP workflow is acquisition provenance only. It is not evidence that the

original device user rooted the phone, and no finding in this report depends on that inference.

2.2 Tools, queries and timestamp method

The examination was conducted in a Kali Linux working environment. SQLite databases were inspected using DB Browser for SQLite, and SQL queries were used to review relevant tables and records. SHA-256 hashes were calculated with sha256sum. The queries and timestamp-validation approach are documented in Appendices B and D. The DB Browser for SQLite version should be added if available. No forensic-suite or parser output is relied upon in this report.

Timestamp formats were determined from each source and the raw value is preserved in the tables. Most Android and application tables store Unix epoch milliseconds (converted by dividing by 1,000). The MMS table pdu.date stores epoch seconds. Google Photos local_media.capture_timestamp stores local wall-clock time encoded as if it were UTC; the true UTC value is utc_timestamp, and the difference equals the stored timezone_offset (-18,000,000 ms). Google Maps coordinates in gmm_myplaces and gmm_sync are E6 microdegrees (divide by 1,000,000). The accounts_de debug_table writes formatted date strings in device local time after initial setup. All converted times below are labelled UTC, with device-local time (UTC-05:00) shown where useful. Examiner-computer local time was never used.

Time zone evidence. Twenty-six of 27 Google Photos rows carry timezone_offset -18,000,000 ms (UTC-05:00) and one September 2019 row carries -14,400,000 ms (UTC-04:00). That pattern matches US Eastern daylight time in September and standard time in January to February, but it states the zone recorded by the app, not a user setting. The accounts_de debug_table shows the same five-hour difference from accounts_de timestamps for accounts added after initial setup (for example the TextNow account: 14:54:52 in debug_table against 19:54:52 UTC). The earliest Google account rows show no offset, so the device clock or zone was not yet applied during first setup. Time zone is therefore treated as UTC-05:00 with that qualification.

2.3 Items listed in the brief but not supplied

The following sources were not in the archive and are not examined: /product/build.prop and

/system/build.prop; userlist.xml and per-user XML; packages.list, packages.xml and package-usage.list;

/data/system/usagestats; WifiConfigStore.xml; settings_secure.xml; Gmail and other shared

preferences; the Kik message database; and application media files. Findings in those families are recorded as absent from the supplied material rather than as absent from the device.

3 Device, account and system profile

3.1Build and device identity

No build properties were supplied, so model, codename, SDK, build number, security patch and serial number are not reported. The Pixel 3 / Android

10 identity comes from the brief, not from an examined file. The only device-level values in the supplied databases are gservices.db main.device_country = us and the Google Fi carrier entries in telephony.db.

3.2 Users and owner evidence

locksettings.db holds rows for user IDs 0 and 11 (plus a global row, user -2). Without userlist.xml the nature of user 11 (secondary, managed or profile user) is unverified. profile.db contains one

owner-profile raw contact with a display name and one account row with no name or type. contacts2.db and accounts_ce.db are consistent with a single primary Google account that also owns the calendar.

Owner evidence shows a configured profile, not who used the device at any time.

3.3 Accounts

accounts_ce.db and accounts_de.db each hold 13 accounts with matching identifiers. One is a Google account and twelve are application accounts. Account names are masked below. Credential material (authtokens: 105 rows; extras: 267 rows) was not read or reported. In accounts_de.accounts the column last_password_entry_time_millis_epoch coincides with the add time recorded in debug_table, so it is read as the account creation or last sign-in time rather than a password event.

ID	Account type (package)	Name (masked)	Raw epoch ms	UTC
1	com.google	th***@gmail.com	1580312467984	2020-01-29 15:41:07
2	org.telegram.messenger	85****690	1580322664981	2020-01-29 18:31:04
3	org.thoughtcrime.securesms	Si***	1580322700050	2020-01-29 18:31:40
4	com.imo.android.imous	im***	1580322720395	2020-01-29 18:32:00
5	com.zhiliaoapp.musically	Ti***	1580323396517	2020-01-29 18:43:16
6	com.facebook.messenger	Me***	1580323862026	2020-01-29 18:51:02
7	com.twitter.android.auth.login	TD***	1580326971888	2020-01-29 19:42:51
8	com.whatsapp	Wh***	1580332202948	2020-01-29 21:10:02
10	com.silentcircle.account	th***	1580589691083	2020-02-01 20:41:31
11	com.skype.raider	Sk***	1580591300780	2020-02-01 21:08:20
12	com.enflick.android.TextNow.account	Te***	1581191692330	2020-02-08 19:54:52
13	com.viber.voip	+1****674	1581192938951	2020-02-08 20:15:38
14	com.google.android.apps.tachyon	Duo	1581643155834	2020-02-14 01:19:15

accounts_de.meta maps account types to the owning app UID, giving UID 10244 for com.whatsapp, 10239 for TextNow, 10231 for Viber and 10141 for the Google account type. An account registered with the system shows that an app asked for one; it does not show how the app was used.

4.0 SIM, carrier and connectivity

siminfo row	Slot / state	Carrier (MCC/MNC)	Number (masked)	ICC ID (masked)
_id 1	sim_id -1 (not in a slot at acquisition)	Google Fi (310/260)	ends 1146	ends 6693
_id 2	sim_id 0 (slot 0)	Google Fi (310/120)	+1 ... ends 4674	ends 3842

These are subscriber and network values (SIM profile, carrier, country code us); they are separate from the account records in 4.3 and from call or message data. The carriers table lists 16 Google Fi carrier and APN entries, which are configuration, not usage. The call log links calls 1 to 6 (31 January to 8 February) to the ICC ID of siminfo _id 1 and calls 7, 9 and 11 to 14 (12 to 14 February) to _id 2 (calls 8 and 10 carry no subscription value), which indicates a change of active profile between 8 and 12 February. Bluetooth and saved Wi-Fi data were not supplied and are not examined.

4.1 System and Google service settings

locksettings.db shows lockscreen.disabled = 0 for user 0, and lockscreen.password_type 196608 for user 0 and 131072 for user 11. Read against the AOSP quality constants these correspond to complex-numeric and numeric credential classes; that interpretation should be checked against the Android 10 source. Credential hashes and password history were not read. The row only states that a lock credential type was configured. It does not show when it was set or used. googlesettings.db (partner table, 9 rows) records network_location_opt_in = 1 and use_location_for_services = 1, which are settings and do not show that any location was collected. gservices.db main (1,914 rows) contains device_country = us and check-in policy keys; saved_global, saved_secure and saved_system hold default entries.

4.2 Application inventory

localappstate.db (Google Play, appstate table) holds 94 package records: 63 Google, AOSP or Mobile Services packages (prefixes com.google., com.android., android.) and 31 third-party packages. Thirty of the third-party records carry first_download_ms values between 1580317327052 and 1580317923118 (29 January 2020, 17:02:07 to 17:12:03 UTC), a ten-minute block that suggests batch installation.
Examples: TextNow (17:08:03 UTC), Kik (17:08:43 UTC) and WhatsApp (1580317744179, 17:09:04 UTC). The clustering shows when Play recorded the first download; it does not show who initiated it or why. packages.xml was not supplied, so installer, code path and update times are not reported.

4.3 Usage evidence

usagestats and package-usage.list were not supplied and no parser was run, so there are no foreground, background or interaction events to report. Application-database timestamps in section 5 record messages, calls or media and must not be read as usage events.

4.4 Cross-source relationship: WhatsApp

Three independent sources order the WhatsApp onboarding. Play first downloaded com.whatsapp at 2020-01-29 17:09:04 UTC (localappstate.db). A WhatsApp verification SMS arrived at 18:37:07 UTC (mmssms.db sms _id 8, date_sent 1580323027000; mirrored in bugle_db messages _id 8). The WhatsApp account was registered with the system at 21:10:02 UTC (accounts_de.db accounts _id 8, 1580332202948). The sequence of install, verification message and account creation is consistent, but the 2 h 33 min gap between the SMS and account creation, and the intervening activity for

other apps, mean it does not by itself describe a single sitting.

5 Artefact findings

5.1 AOSP communications, contacts, calls and calendar

SMS (mmssms.db, sms). The table has 24 rows: 22 inbound (type 1; 20 flagged unread) and 2 outbound (type 2). Most inbound rows come from short codes and carry app verification codes between 29 January and 1 February, with Facebook notification messages recurring from 3 to 13 February. Verification codes are not reproduced here. The two outbound messages go to the number ending 0479 (Contact A): sms _id 21 at 1581273702392 (2020-02-09 18:41:42 UTC) and _id 25 at 1581685743277
(2020-02-14 13:09:03 UTC). Outbound rows have date_sent = 0, so only the stored date is available.
MMS (pdu, part, addr). Seven pdu rows are linked to ten part rows through part.mid. Three of the parts are SMIL layout files, so there are seven messages and seven text parts, not ten. Three are outbound (msg_box 2, m_type 128: 1 February 20:04:53, 20:07:52 and 20:09:27 UTC) and four are inbound (msg_box 1, m_type 132: 1 February 20:06:04 and 20:08:12; 9 February 18:43:08; 14 February 13:10:08 UTC). All seven are in thread 17 with the number ending 0479, and pdu.date is stored in seconds. Text parts are short conversational lines; no media parts are present.
Contacts and calls. contacts2.db has 8 account rows, 10 raw contacts, 1 aggregated contact and 9 rows in deleted_contacts (deletion times 1580507693823 to 1581644288610 ms). All raw contacts carry the same display name, and phone_v2 rows give the number ending 0479 in formats such as (919) 579-0479 and 9195790479 (two rows hold a non-standard +99-prefixed value), so the address book labels that number as one individual (Contact A). A label shows how the address book names a number; it does not show who held the phone at a given time. calllog.db holds 14 calls: 2 incoming, 2 outgoing, 8 missed and 2 voicemail-type (type 4). The first five missed calls have no stored number.

Calls 11 to 14 (14 February 01:29:01, 01:30:24, 13:05:57 and 14:41:18 UTC) involve Contact A, with
durations of 11, 65, 86 and 91 s. The two type-4 rows (3 s each) follow missed calls 7 and 9 by 51 s and 49 s, consistent with voicemail notifications; voicemail.db itself is empty and voicemail_status shows the Dialer as configured.
Calendar. calendar.db contains one Google calendar in Calendars and sync metadata, colour and cache rows, but Events, Attendees, Reminders and Instances each have zero rows. This is a documented empty calendar result: there is no organiser evidence in the supplied database.

5.2 Google applications, media and location

Google Messages (bugle_db). messages holds 35 rows across 14 conversations. Twenty-three rows have message_protocol 0 and an sms_message_uri of the form content://sms/N, eight have protocol 3 (seven with content://mms/N), and four have protocol 5 with no provider URI (status 210, created after inbound short-code messages). Protocol meanings are inferred from the URI linkage and were not confirmed against the app. The URI values tie records to the Telephony Provider: for example bugle messages _id 33 is content://sms/21 (received_timestamp 1581273702392 equals sms.date) and _id 34 is content://mms/6. No row in the examined values identifies RCS, but RCS cannot be excluded from that alone.
Google Dialer. annotated_call_log.db holds 12 rows. Row identifiers, timestamps, types and durations match calllog.db for _ids 1 to 7, 9 and 11 to 14. The two voicemail-type rows (_ids 8 and 10) are present in the provider but absent from the annotated log. The number is stored as an encoded blob (a

length-prefixed number string, a country code US and a flag), which decodes to the same numbers as calllog.db. dialer.db holds smart-dial entries (2 rows) only, and voicemail.db has no data.

Google Maps. gmm_myplaces.db has one synced item with E6 coordinates 35734602 / -78636654 (35.735, -78.637) and timestamp 1554407343604 (2019-04-04 19:49:03 UTC), which pre-dates the January to February 2020 activity by about nine months. gmm_sync.db has 12 corpus metadata and 7 item rows, and gmm_storage.db has 4 serialised entries including a CompletedNavigation record. The serialised blobs were not decoded in this draft, so no route or navigation content is reported. A saved place or sync item does not show that a person was at that location.

Google Photos (gphotos-1.db). local_media has 27 rows (media: 26), none deleted; local_trash.db contains no records; auto_backup_state is 0 for all local rows; 21 local rows (20 in media) carry is_hidden = 1, a flag whose meaning in this app version was not verified. Most files come from messaging and social app folders (Viber, TextNow, WhatsApp, Telegram, Twitter, Imgur and others). Only three rows carry coordinates: IMG_20190917_082637.jpg (Restored; 35.7356 / -78.6364, offset UTC-04:00), IMG_20200209_132353.jpg (35.661 / -78.880, 9 February 18:23:53 UTC) and

IMG_20200213_195030.jpg (35.659 / -78.874, 14 February 00:50:31 UTC). The 2019 photograph lies about 113 m from the Maps saved place; both pre-date the window and neither shows physical presence.

5.3 Third-party applications

TextNow (textnow_data.db). One conversation with a number ending 2223 has nine message rows between 1581192046000 and 1581192670000 ms (2020-02-08 20:00:46 to 20:11:10 UTC): three

inbound texts, an inbound missed-call notice (stored as a text row), an inbound image, an outbound text, an outbound image and two outbound rows with message_type 102 and 201 (meaning unverified). The image rows give local attachment paths under /storage/emulated/0/ and a media URL whose token was not decoded. The contacts table has 15 entries, including one username entry and one support address. The messages show app activity on the account; they do not identify who typed them.

WhatsApp (msgstore.db; wa.db). msgstore.db has 19 message rows: seven placeholder rows (one with timestamp 0 and six status@broadcast rows with February 2017 values, treated as

application-seeded), three rows in a separate chat on 29 January at 18:50 UTC, and nine rows in a chat with the number ending 0479. That chat runs from 2020-02-08 20:55:14 to 2020-02-09 18:05:02 UTC: text exchanges, an inbound image (message_media path Media/WhatsApp Images/IMG-20200208-WA0000.jpg), an outbound image (IMG-20200208-WA0001.jpg), two

live-location messages (media_wa_type 16) and a location message (type 5). call_log has two calls: an inbound video call (timestamp 1581195562000, 86 s) and an outbound voice call (1581195700411, 71

s), both with call_result 5, a code whose meaning was not verified. wa.db is a 4,096-byte file with no tables, so no WhatsApp contact data is available.

Kik (kikCoreDatabase.db). CoreTable has one row with is_active = 1 and a username. This is account context only. No conversation database was supplied, so Kik messages cannot be assessed. Two applications (TextNow, WhatsApp) therefore provide attributable activity, and Kik's state is documented as account-only.

6 Event chronology

Window and justification. The window is 2020-02-08 19:30:00 UTC to 2020-02-09 19:30:00 UTC (24 hours; 14:30 to 14:30 device-local time, UTC-05:00). It was chosen because it holds the first day on which TextNow, Viber and WhatsApp records all appear together, the first conversation with the same counterpart number in WhatsApp, and the next day's location sharing, photograph and SMS/MMS exchange. It therefore covers eight databases and supports several
cross-source checks. Each time is converted from the raw value shown.

#	Raw value (unit)	Converted (UTC)	Local UTC-05:00	Event	Source (database / table / record)	Qualification
E01	1581191692330 (ms)	2020-02-08 19:54:52	14:54:52	TextNow account registered with AccountManager	accounts_de.db / accounts _id 12; debug_table add at 14:54:52 local	Account registration only; not the operator
E02	1581192046000 (ms)	2020-02-08 20:00:46	15:00:46	First of 9 TextNow rows (20:00:46 to 20:11:10 UTC) with counterpart ending 2223	textnow_data.db / messages _id 16 (to _id 25)	App record; sender identity not shown
E03	1581192896305 (ms)	2020-02-08 20:14:56	15:14:56	Missed call from a +33 number	calllog.db / calls _id 6; annotated_call_log.db _id 6	Logged against SIM profile 1; caller unknown
E04	1581192938951 (ms)	2020-02-08 20:15:38	15:15:38	Viber account registered (name equals line-2 number)	accounts_de.db / accounts _id 13; debug_table 15:15:38 local	Follows E03 by 42 s; link not established (see 7)
E05	1581193451000 (ms)	2020-02-08 20:24:11	15:24:11	Viber image file recorded in Google Photos	gphotos-1.db / local_media _id 23 (utc_timestamp)	File metadata; not send or receive time
E06	1581194137000 (ms)	2020-02-08 20:35:37	15:35:37	TextNow image file recorded in Google Photos	gphotos-1.db / local_media _id 22; textnow_data messages _id 24 attach path	Message dated 20:09:16 UTC; file 26 min later
E07	1581195314692 (ms)	2020-02-08 20:55:14	15:55:14	WhatsApp outbound text to number ending 0479	msgstore.db / messages _id 12 (key_from_me 1)	App record; sender not shown
E08	1581195365000 (ms)	2020-02-08 20:56:05	15:56:05	WhatsApp inbound text reply	msgstore.db / messages _id 13	App record
E09	1581195425000 (ms)	2020-02-08 20:57:05	15:57:05	WhatsApp inbound image (saved as IMG-20200208-WA0000.jpg)	msgstore.db / messages _id 14, message_media; gphotos local_media _id 21 at 20:57:08	Corroborated by Photos record 3 s later
E10	1581195492693 (ms)	2020-02-08 20:58:12	15:58:12	WhatsApp outbound image (IMG-20200208-WA0001.jpg)	msgstore.db / messages _id 16, message_media	No matching Photos row
E11	1581195562000 (ms)	2020-02-08 20:59:22	15:59:22	WhatsApp inbound video call, 86 s	msgstore.db / call_log _id 1	call_result 5 meaning unverified
E12	1581195700411 (ms)	2020-02-08 21:01:40	16:01:40	WhatsApp outbound voice call, 71 s	msgstore.db / call_log _id 2	As above
E13	1581270953000 (ms)	2020-02-09 17:55:53	12:55:53	Camera video recorded (no coordinates)	gphotos-1.db / local_media _id 20	Metadata only

#	Raw value (unit)	Converted (UTC)	Local UTC-05:00	Event	Source (database / table / record)	Qualification
E14	1581271214337 (ms)	2020-02-09 18:00:14	13:00:14	WhatsApp live-location message (first of two, 18:00:14 and 18:00:33)	msgstore.db / messages _id 17 (type 16), _id 18	Coordinates 35.661 / -78.880 (rounded)
E15	1581271502890 (ms)	2020-02-09 18:05:02	13:05:02	WhatsApp location message	msgstore.db / messages _id 19 (type 5)	Device receipt 18:07:24; read 18:15:25
E16	1581272633139 (ms)	2020-02-09 18:23:53	13:23:53	Geotagged camera photo IMG_20200209_132353.jpg	gphotos-1.db / local_media _id 24	9.5 m from E15 location; not proof of presence
E17	1581273702392 (ms)	2020-02-09 18:41:42	13:41:42	Outbound SMS to number ending 0479	mmssms.db / sms _id 21; bugle_db messages _id 33 (content://sms/21)	Presentation layer mirrors provider
E18	1581273788 (s)	2020-02-09 18:43:08	13:43:08	Inbound MMS reply (short text) from 0479	mmssms.db / pdu _id 6, part _id 9; bugle_db messages _id 34	pdu.date in seconds

ID	Relationship	Sources	Result
R1	WhatsApp inbound image E09 and the Google Photos file IMG-20200208-WA0000.jpg	msgstore.db (messages, message_media) and gphotos-1.db (local_media)	Corroborated: same file name; Photos record 3 s after the message. Independent databases.
R2	WhatsApp location message E15 and geotagged photo E16	msgstore.db and gphotos-1.db	Corroborated: positions 9.5 m apart, 19 min 51 s apart. Shows device-side records of one place, not who was there.
R3	Outbound SMS E17 and inbound MMS E18 in the Telephony Provider and Google Messages	mmssms.db and bugle_db	Linked by content:// URIs and matching timestamps. Same underlying messages, so corroborates existence and time, not independence of origin.
R4	TextNow attachment path and the Photos TextNow Images file (E02 and E06)	textnow_data.db and gphotos-1.db	Partly corroborated: same file name; Photos time is 26 min later than the message time, unresolved.
R5	Missed call E03 and Viber account E04	calllog.db / annotated_call_log.db and accounts_de.db	Qualified: 42 s apart, but call is logged on SIM profile 1 while the Viber account name matches the profile-2 number.

7 Interpretation and limits

7.1 What the evidence supports

F1. Device and account context. The databases show one Google account and twelve application accounts registered on the device between 29 January and 14 February 2020, two Google Fi SIM profiles with a change of active profile between 8 and 12 February, a configured lock credential, a batch Play installation of about 30 third-party apps on 29 January, and a device-local time zone of UTC-05:00.

Alternative explanations: batch installation and bulk account creation are consistent with scripted training data generation as well as with ordinary setup, and the evidence cannot separate them. No build, user-list or Wi-Fi findings are made because those sources were not supplied.

F2. System or application use. App records (messages, calls, media files) show that TextNow, WhatsApp, Viber-related media and Google apps held data in the 8 and 9 February window. They do not show foreground use, because no usagestats data was supplied. Application presence, an installation record and an account row do not identify the person operating the device.

F3. Communications and organiser activity. The Telephony Provider, Google Messages and Dialer agree on a conversation with the number ending 0479 (SMS, seven MMS, calls) and on mostly automated inbound short-code messages. WhatsApp and TextNow add separate conversations and calls. The calendar is empty. The contact label is an address-book name, not proof of who held that number. Differences exist: the Dialer annotated log omits the two voicemail-type rows and wa.db is empty, so some expected context is absent.

F4. Reconstructed chronology. Eighteen events from eight databases fit a coherent 24-hour sequence: TextNow activity and a Viber registration on the evening of 8 February, a WhatsApp conversation with an image and two calls shortly before 21:02 UTC, then a location share, a geotagged photograph and an SMS/MMS exchange on 9 February between 17:55 and 18:44 UTC. Two relationships (R1, R2) are corroborated by independent databases. Times depend on the UTC-05:00 assumption only for the local-time column; UTC values come directly from raw data.

7.2 Contradictions and uncertainty

Four points remain unresolved. First, the missed call on 8 February is logged against the first SIM profile while the Viber account is named for the second profile's number, so the apparent link between them (42 s apart) is not established. Second, the TextNow image file appears in Photos 26 minutes after its message time; the file may have been saved or opened later, or the message time may be a server value. Third, WhatsApp call_result 5, the TextNow message types 102 and 201, Google Messages protocol 5 and the is_hidden flag were not confirmed against application documentation. Fourth, the Maps serialised blobs were not decoded, so navigation content is unassessed. Each affects the weight of the related findings and none is resolved by guesswork in this report.

7.3 Limits of scope

The evidence is an extract of 25 databases from a logical copy taken after TWRP root. Deleted data outside those files, files not supplied (section 3.4) and encrypted content are outside the analysis. No conclusion is drawn about any person's conduct, identity or intent, and references to Contact A and number endings are for traceability only.

8 Appendices

8.1 Appendix A Finding log

ID	Finding	Source (path / database / table / record)	Raw time or value	Converted time	Qualification
F-01	Device-local time zone UTC-05:00 (UTC-04:00 for one 2019 row)	Task D/gphotos-1.db / local_media.timezone_offset	-18000000 (26 rows); -14400000 (1 row)	UTC-05:00 / UTC-04:00	App-recorded zone; first-setup rows show no offset
F-02	Google account added	Task A/accounts_de.db / accounts _id 1	1580312467984	2020-01-29 15:41:07 UTC	Registration only; credentials not read
F-03	13 accounts (1 Google, 12 app) in both CE and DE databases	Task A/accounts_ce.db and accounts_de.db / accounts	13 rows each	29 Jan to 14 Feb 2020	Account presence is not use
F-04	Two Google Fi SIM profiles; active profile changes in call log	Task A/telephony.db / siminfo; Task C/calllog.db / calls.subscription_id	siminfo _id 1, _id 2	Calls 1-6 profile 1; calls 7, 9, 11-14 profile 2	Subscriber data; not usage
F-05	Lock credential configured (user 0 type 196608; user 11 type 131072)	Task B/locksettings.db / locksettings	196608; 131072	Not time-stamped	Type only; hashes not read; constants to verify
F-06	Batch Play installation of third-party apps	Task B/localappstate.db / appstate.first_download_ms	1580317327052 to 1580317923118	2020-01-29 17:02:07 to 17:12:03 UTC	Play record; initiator unknown
F-07	WhatsApp install, SMS code and account creation in sequence	localappstate.db; Task C/mmssms.db sms _id 8; accounts_de.db _id 8	1580317744179; 1580323027000; 1580332202948	17:09:04; 18:37:07; 21:10:02 UTC (29 Jan)	Consistent order; not one sitting
F-08	SMS: 24 rows (22 in, 2 out)	Task C/mmssms.db / sms	date ms; _id 21 = 1581273702392	_id 21: 2020-02-09 18:41:42 UTC	Codes not reproduced
F-09	MMS: 7 messages from 10 parts (no double count)	Task C/mmssms.db / pdu, part, addr	pdu.date seconds; _id 6 = 1581273788	_id 6: 2020-02-09 18:43:08 UTC	SMIL parts excluded
F-10	Call log: 14 calls incl. 2 voicemail-type	Task C/calllog.db / calls	e.g. _id 11 = 1581643741051	2020-02-14 01:29:01 UTC	Name is address-book label
F-11	Calendar empty (0 events, attendees, reminders)	Task C/calendar.db / Events, Attendees, Reminders	0 rows	n/a	Documented empty result
F-12	Google Messages rows link to provider URIs	Task D/bugle_db / messages (sms_message_uri)	_id 33 = content://sms/21	2020-02-09 18:41:42 UTC	Protocol values inferred
F-13	Dialer log omits two voicemail-type rows	Task D/annotated_call_log.db vs calllog.db	12 vs 14 rows	n/a	Reason not established
F-14	Maps saved place (E6)	Task D/gmm_myplaces.db / sync_item	35734602 / -78636654; 1554407343604	35.735 / -78.637; 2019-04-04 19:49:03 UTC	Saved place is not presence
F-15	WhatsApp location and geotagged photo 9.5 m apart	msgstore.db messages _id 19; gphotos-1.db local_media _id 24	1581271502890; 1581272633139	18:05:02 and 18:23:53 UTC (9 Feb)	Independent DBs; not identity

ID	Finding	Source (path / database / table / record)	Raw time or value	Converted time	Qualification
F-16	WhatsApp image corroborated in Photos	msgstore.db messages _id 14; gphotos-1.db local_media _id 21	1581195425000; 1581195428000	20:57:05 and 20:57:08 UTC (8 Feb)	3 s apart
F-17	WhatsApp calls (86 s video in; 71 s voice out)	msgstore.db / call_log _id 1, 2	1581195562000; 1581195700411	20:59:22; 21:01:40 UTC (8 Feb)	call_result 5 unverified
F-18	TextNow conversation, 9 rows	Task D/textnow_data.db / messages _id 16-25	1581192046000 to 1581192670000	20:00:46 to 20:11:10 UTC (8 Feb)	Sender not shown
F-19	Kik account row only; wa.db empty	Task D/kikCoreDatabase.db CoreTable; Task D/wa.db	1 row; 0 tables	n/a	No Kik messages supplied
F-20	24-hour chronology with 18 events	Section 6	see Section 6	2020-02-08 19:54:52 to 2020-02-09 18:43:08 UTC	Window justified in Section 6

Appendix B Queries used (read-only)

```
-- Accounts (names/types only; credentials not selected)
SELECT _id,name,type,last_password_entry_time_millis_epoch,datetime(last_password_entry_time_millis_epoch/1000,'unixepoch') AS utc FROM accounts;
-- Play first download
SELECT package_name,first_download_ms,datetime(first_download_ms/1000,'unixepoch') AS utc FROM appstate ORDER BY first_download_ms;

SELECT _id,thread_id,address,type,read,date,datetime(date/1000,'unixepoch') AS utc FROM sms ORDER BY date;
-- SMS
-- MMS: join pdu and part by mid (dates in seconds)
SELECT p._id,datetime(p.date,'unixepoch') AS utc,p.msg_box,p.m_type,pt.ct,pt.text FROM pdu p JOIN part pt ON pt.mid=p._id ORDER BY p.date;

SELECT _id,number,type,date,datetime(date/1000,'unixepoch') AS utc,duration,subscription_id FROM calls ORDER BY date;
-- Call log
-- Photos: capture_timestamp is local-as-UTC; use utc_timestamp
SELECT _id,filename,folder_name,utc_timestamp,datetime(utc_timestamp/1000,'unixepoch') AS utc,timezone_offset,latitude,longitude FROM local_media ORDER BY utc_timestamp;

SELECT _id,key_remote_jid,key_from_me,media_wa_type,latitude,longitude,timestamp,datetime(timestamp/1000,'unixepoch') AS utc FROM messages ORDER BY timestamp;
SELECT _id,from_me,video_call,duration,call_result,timestamp,datetime(timestamp/1000,'unixepoch') AS utc FROM call_log;
-- WhatsApp
-- TextNow
SELECT _id,message_direction,message_type,date,datetime(date/1000,'unixepoch') AS utc FROM messages ORDER BY date;
```

All queries use UTC ('unixepoch'), never 'localtime'. Opened with: sqlite3.connect('file:PATH?mode=ro&immutable=1', uri=True).

Appendix C Source paths

Family	Supplied path	Brief Appendix A location (unverified)
Accounts	Task A/accounts_ce.db; accounts_de.db	/data/system_ce/0/ and /data/system_de/0/
Carrier	Task A/telephony.db	/data/data/com.android.providers.telephony
Google settings	Task B/googlesettings.db; gservices.db	/data/data/com.google.android.gsf/databases/
Play state / lock	Task B/localappstate.db; locksettings.db	Play Store databases; /data/system/locksettings.db
AOSP telephony and contacts	Task C/mmssms.db; contacts2.db; calllog.db; profile.db	/data/data/com.android.providers.telephony and .contacts
Calendar	Task C/calendar.db	/data/data/com.android.providers.calendar/databases/

Family	Supplied path	Brief Appendix A location (unverified)
Google Messages / Dialer	Task D/bugle_db; annotated_call_log.db; dialer.db; voicemail.db	com.google.android.apps.messaging; com.google.android.dialer
Maps / Photos	Task D/gmm_*.db; gphotos-1.db; local_trash.db	com.google.android.apps.maps; com.google.android.apps.photos
Third party	Task D/textnow_data.db; msgstore.db; wa.db; kikCoreDatabase.db	com.enflick.android.textnow; com.whatsapp; com.kik.android

Appendix D Manual validation

V-01 Epoch conversion. All 18 raw timestamps used in the chronology (E01 to E18) were converted twice: by SQLite datetime(x/1000,'unixepoch') (seconds for E18) and by an independent Python datetime conversion. All 18 values matched to the second. **V-02 Photos time zone.** The file name IMG_20200209_132353.jpg encodes 13:23:53 local time; the same record's utc_timestamp (1581272633139) converts to 18:23:53 UTC, a five-hour difference equal to timezone_offset, so the manual result matched the parsed field. The TextNow file name image-2020-02-08-15-35-37.jpg and its utc_timestamp (20:35:37 UTC) matched in the same way. No forensic-suite or parser output was used, so these checks compare database fields with independently derived values.

Appendix F Evidence log and file hashes

Field	Entry
Case identifier	CIPB106-Android10-C11_26_DFIT_17305
Source	db.zip supplied by examiner (25 extracted databases)
Archive SHA-256	0ce7de66942ce4f40012b0efd6a2948fb9139c9c287e2659dcc0b43cf15a25f1
Expected package SHA-256 (brief)	ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a (applies to the complete package; not comparable to this archive)
Date and time accessed	
Working copy	Extracted copy of db.zip; databases opened read-only and immutable
Tools	Python 3 sqlite3; sha256sum; unzip.
Time conversion methods	Epoch ms / 1000; pdu.date seconds; Photos utc_timestamp; E6 / 1,000,000; no examiner local time
Sources not supplied	build.prop, userlist.xml, packages.xml, usagestats, WifiConfigStore.xml, settings_secure.xml, Kik message database

File (as supplied)	Bytes	SHA-256
Task A/accounts_ce.db	266240	aeecec36280fb8156c1669f0fd06d0e3e621740699a56a38affdd38630b26c0f
Task A/accounts_de.db	65536	b436a39e5f7c6f56fcb93715c85454f93eb07866699ac34cbb874737935c9ff5
Task A/telephony.db	28672	39c61df8899ea1ddfb7e457b4655ea343c17039078a81dfd219ab750a4f959c9
Task B/googlesettings.db	28672	58da18c17d6169291 64f82bb7f171387 3fb7f7db012901c3236f696f5bd783bf
Task B/gservices.db	323584	20a741e1f9b85937c7b23f078aa22de442ffe012fa0f17c105fce785132b8116
Task B/localappstate.db	376832	9b393a61f4911e5f60651d5341ba0c7fa1317918 23c8d2cbd8c9ec74539b92ad
Task B/locksettings.db	20480	b8b15bdccfee2d678365180c6c7731fd7b9d7270e2059d990857e72dbbceb9ff
Task C/calendar.db	122880	82a580901fca7580dd8841a1c988dcaa65afef068f386d603aeec6039e5aca3c
Task C/calllog.db	40960	5979a3fa0b4e70aff6dc20675e84a14fa584e52ad4d224d9da14d8aa0ef809d3
Task C/contacts2.db	376832	2a1fee3d9412e2f746592fcc0f7abcdd0ece26bc02c95240fd210464f4625050
Task C/mmssms.db	118784	cd5452f5293325 2f9dca917 7491e0a316b542755ac7eb40e61b4b5e3c53f1342
Task C/profile.db	376832	879afe1152fd6a2e9e04d51b4cb332d73514d12d5b8ee8c213b932 1865b6b677
Task D/annotated_call_log.db	24576	e7b896b11cdc375b938fba749ab9d2fab1aeec408a88c2b0ff50cb6f6a417e59
Task D/bugle_db	425984	ed4f19eb9fb60fea7 2545006be9cd6b47171cb88228 94f7da3276a57ea7e3a33
Task D/dialer.db	77824	69d2c0f4329c9c8ed401aebe5cec6cd3652fcd008f2488618781a282f7eb3113
Task D/gmm_myplaces.db	40960	9d09ce47a0c03bd5ffaad8a3869e0852d430e915dad852d08855b7a8ca47f7d8
Task D/gmm_storage.db	94208	051f30146570a091b801d15b46dde4c8a84fcfbb9ce26e922d48b1c98efe052c
Task D/gmm_sync.db	45056	2e96f7a1ab550b26b13675ca972f5595 40c955f52b6e55dd4c71de39d2f59ce2
Task D/gphotos-1.db	1232896	55caad23f7bc42d944a4fb61f0fe831373 30f800b00a5538c449de85a8331e20
Task D/kikCoreDatabase.db	20480	e334661c15d1b325c3f30f69557e2b03b60aa6bf5341bae8ffd9ba55c6923f0b
Task D/local_trash.db	28672	61ba4d2099a5447b186dc839abb9ef7160b5144d471f4e7601c7a7370d89991e

File (as supplied)	Bytes	SHA-256
Task D/msgstore.db	823296	9e133d7262f526b1dab7313f636a1a3f32984d8008310a1a699c83496dd13105
Task D/textnow_data.db	155648	3ceb390eb1a197c38c1f44772bbce80652731319da109bc6f6b3aeddc75edc86
Task D/voicemail.db	24576	760c5ff7f09f76daf06cf70564b8e5449b091d1cff06b5e0cf3027a1dbc39141
File (as supplied)	**Bytes**	**SHA-256**
Task D/wa.db	4096	0a8ffb6b3279635587 16e87db894601 6d143e39f895fa1b43e95ba7032ce2514

Appendix G Academic integrity declaration

I confirm that this report reflects my own examination of the authorised course evidence and that I have identified all sources, tools, and assistance used.

Sources, tools and assistance used: course brief and Android course slides; Kali Linux; DB Browser for SQLite; sha256sum; and any other tools actually used during the examination. Add the DB Browser for SQLite version if known. Disclose any AI assistance accurately and in accordance with course policy. Confirm that the queries and findings have been independently re-run and verified before submission.

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